

Research on Inventory Management of Medical Supplies Based on Hybrid Intelligent Optimization Algorithm

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Abstract—Effective inventory management of medical supplies is vital in healthcare due to fluctuating demand, strict expiration dates, and the need for multi-objective optimization. Traditional methods often struggle with variable demand and complex constraints, resulting in excess inventory or shortages. This article introduces GA-PSO, a hybrid optimization algorithm that combines genetic algorithms with particle swarm optimization. By integrating dynamic demand forecasting and multi-objective optimization, GA-PSO enhances medical supply management. Simulation results indicate that GA-PSO significantly outperforms traditional methods, achieving a 39.3% reduction in costs, a stock-out rate of just 1.3%, and minimal shelf life loss at 0.6%. This study offers an intelligent decision-making tool for medical supply inventory, greatly improving operational efficiency and service quality while providing insights for smart healthcare supply chains.

Keywords—Management of medical supplies inventory; Hybrid intelligent optimization algorithm; Genetic algorithm; Particle swarm optimization; Multi-Objective Optimization

I. INTRODUCTION

Efficient management of medical supply inventory is crucial for the supply chain, characterized by fluctuating demand, expiration risks, diverse product categories, and stringent regulations. High-value consumables necessitate meticulous monitoring of temperature and humidity, while public health emergencies can trigger demand surges or shortages, complicating inventory management. Traditional approaches, such as economic order quantity (EOQ), struggle to keep pace with these dynamic requirements, frequently leading to overstock situations, elevated costs, or material shortages. Inadequate inventory management can result in an 8% to 12% loss rate among leading hospitals and contribute to over 15% of service delays attributable to shortages.

Recent developments in optimization algorithms and technologies—such as machine learning for demand forecasting, Internet of Things (IoT) for real-time monitoring, and blockchain for traceability—present promising solutions. However, standalone algorithms encounter difficulties in multi-objective optimization: genetic algorithms (GA) are effective for global searches but often converge slowly, while particle swarm optimization (PSO) is proficient locally but may overlook more comprehensive solutions. Moreover, existing systems frequently fail to meet the unique demands of the medical device sector, including unique device identification (UDI) and cold chain logistics.

This research proposes a hybrid GA-PSO algorithm designed to optimize medical supplies inventory more effectively by addressing local optima and enhancing convergence efficiency. This approach aims to balance inventory costs, stockout risks, and expiration losses. Additionally, the study integrates a Long Short-Term Memory (LSTM) neural network for demand forecasting, establishing a "forecast-optimization-execution" closed-loop system to facilitate more intelligent, data-driven decision-making.[1.2]

II. SINGLE OBJECTIVE OPTIMIZATION MODEL FOR MEDICAL MATERIAL INVENTORY MANAGEMENT

A. Management of medical supplies inventory

Managing medical supply inventories is a complex optimization challenge that requires balancing factors like cost, service levels, and shelf life. However, healthcare facilities often focus primarily on reducing total costs. This emphasis is driven by increasing operational cost pressures; research from the China Hospital Association in 2023 indicates that inventory costs for medical supplies in tertiary hospitals account for 18% to 25% of overall expenses, making cost control crucial. Additionally, in a single-objective framework, other important metrics can be treated as constraints, such as keeping the out-of-stock rate below 5% as a service level requirement and setting limits on inventory turnover time for shelf-life management.[3.4]

This article posits three essential assumptions to simplify the model:

(1) Stable demand distribution: It is assumed that during the optimization cycle, material demand adheres to a normal distribution $N(\mu, \sigma^2)$, calibrated based on historical data;

(2) Linear cost structure: Purchasing expenses are directly proportional to order quantities, whereas storage costs are linearly associated with average inventory levels;

(3) Immediate replenishment mechanism: The variability of supplier delivery cycles is disregarded, with the assumption that replenishment occurs instantaneously upon ordering.

While these simplifications may overlook certain complexities of real-world scenarios, they significantly reduce the computational challenges involved in algorithm resolution. Practical implementations have demonstrated that when the demand fluctuation coefficient (σ/μ) falls below 0.5, the

alignment between the results of the simplified model and those of the multi-objective model can exceed 85%. Moreover, the model retains essential cost components, including explicit procurement costs, storage expenses, and implicit opportunity costs related to stockouts, thereby offering practical and effective inventory decision support for healthcare institutions.

B. Mathematical model construction

1. Objective function (minimizing total cost)

$$\min TC = c_p \cdot Q + c_h \cdot \frac{Q}{2} + c_s \cdot E[\max(D - Q, 0)] \quad (1)$$

Among them,

Procurement Cost ($c_p \cdot Q$): Direct cost proportional to order quantity Q .

Holding Cost ($c_h \cdot \frac{Q}{2}$): Assumes uniform demand consumption; average inventory level is $\frac{Q}{2}$.

Shortage Cost ($c_s \cdot E[\max(D - Q, 0)]$): Assume demand D follows probability density function $f(D)$

c_p is the procurement cost, c_h is the storage cost, and c_s is the out of stock cost, D is the random demand quantity (following the distribution fitted by historical data)

2. Key constraint conditions

(1) Inventory capacity limit:

$$Q \leq Q_{max}$$

The Q_{max} is determined by the physical space of the warehouse.

(2) Validity safety constraints (for drugs/reagents):

$$\frac{Q}{\lambda} \leq T_{safe} - t_{lead}$$

λ is the daily average consumption, T_{safe} is the validity period of the material (days), t_{lead} is the supplier's delivery cycle (days).

(3) Lower limit of service level:

$$P(D \leq Q + R) \geq 90\%$$

III. DESIGN OF HYBRID INTELLIGENT OPTIMIZATION ALGORITHM

This study introduces a hybrid algorithm (GA-PSO) that merges the global search capabilities of genetic algorithms (GA) with the local optimization efficacy of particle swarm optimization (PSO). A multi-tiered optimization framework has been developed to tackle the dynamic constraint attributes inherent in the management of medical material inventories. The algorithm synergizes the preservation of population diversity through genetic operations with the rapid convergence facilitated by particle swarm optimization, thereby enabling effective optimization amidst complex constraints.[5]

A. Encoding and initialization

Here, chromosome/particle encoding is used: each individual represents an inventory strategy, encoded as a three-dimensional vector:[6]

$$X = [Q, R, \alpha]$$

Q is the order quantity (integer code, range: $[0.5D_{avg}, 3D_{avg}]$),

R is the order quantity (integer code, order point (real code, range: $[0, D_{max}]$),

α is the safety stock factor (real number code, range: $[0.2, 1.5]$).

Population initialization: Randomly generate $N=100$ individuals, satisfying the constraint:

$$Q_i \geq \lambda \cdot (t_{safe} + t_{lead})$$

Where λ is the daily average consumption.

B. Design of fitness function

Objective function (minimizing total cost):

$$\min TC = c_p \cdot Q + c_h \cdot \frac{Q}{2} + c_s \cdot E[\max(D - Q, 0)] \quad (2)$$

Constraint handling (penalty function method):

$$Fitness = \frac{1}{TC + \rho \cdot (\sum_{i=1}^3 \max(0, g_i))} \quad (3)$$

Among them, the constraint conditions are:

1. Validity and safety:

$$g_1 = \frac{Q}{\lambda} + t_{lead} - T_{safe}$$

2. Service level:

$$g_2 = 0.9 - P(D \leq Q + R)$$

3. Inventory capacity:

$$g_3 = Q + R - Q_{max}$$

The penalty coefficient $\rho = 10^6$ shows a sharp decrease in fitness when the constraint is violated.

C. genetic manipulation

1. Selection (tournament selection)

Randomly select $k=5$ individuals from the population each time, retain the one with the highest fitness, and repeat until N parent individuals are selected.[6]

2. Intersection (arithmetic intersection)

For the selected parent individual X_a, X_b , Generate offspring:

$$X_c = \gamma X_a + (1 - \gamma) X_b \quad (4)$$

$$X_d = (1 - \gamma) X_a + \gamma X_b \quad (5)$$

Cross probability $P_c = 0.8, \gamma \sim U(0.3, 0.7)$

3. Variation (non-uniform variation)

Perturb each gene locus with a probability of $p_m = 0.05$:

$$x' = x + \Delta(t, x_{max} - x) \cdot \text{sign}(r - 0.5) \quad (6)$$

The disturbance amplitude is:

$$\Delta(t, y) = y \cdot (1 - r^{(1-t/T_{max})^3}) \quad (7)$$

$r \sim U(0, 1)$, T_{max} is the maximum number of iterations.

4. Particle Swarm Optimization Update

1. Speed update equation

For each particle i , update speed:

$$v_i^{t+1} = w \cdot v_i^t + c_1 r_1 (p_{best} - x_i^t) + c_2 r_2 (g_{best} - x_i^t) \quad (8)$$

Among them:

Inertial weight $w = 0.9 - 0.5 \cdot (t/T_{max})$ (linearly decreasing).

Acceleration constant $c_1 = c_2 = 2.0$, $r_1, r_2 \sim U(0, 1)$

2. Position update equation

$$x_i^{t+1} = x_i^t + v_i^{t+1} \quad (9)$$

If the new location exceeds the boundary, use reflective boundary processing:

$$x' = x_{min} + |x - x_{max}| \bmod (x_{max} - x_{min}) \quad (10)$$

5 Mixed Strategy Design

Elite retention mechanism:

Each generation retains the top 10% of individuals with fitness and directly enters the next generation to avoid the loss of high-quality solutions.

Dynamic switching rules:

When the population diversity is $D_{pop} < D_{th}$, enhances the frequency of PSO operation.

When the algorithm falls into a local optimum (with no improvement in the optimal solution for 5 consecutive generations), a GA mutation operation is triggered.

The hybrid intelligent optimization algorithm framework is shown in Figure 1:

D. Parameter Calibration

Parameter setting basis, as shown in Table 1:

TABLE 1 PARAMETER SETTINGS

parameter	values	Calibration method
Population size N	100	Determined through analysis of variance of 30 experiments
Maximum Iteration T_{max}	500	Observe the inflection point of the convergence curve
crossover rate p_c	0.8	Literature comparison and grid search
mutation rate p_m	0.05	The trade-off value for avoiding premature convergence

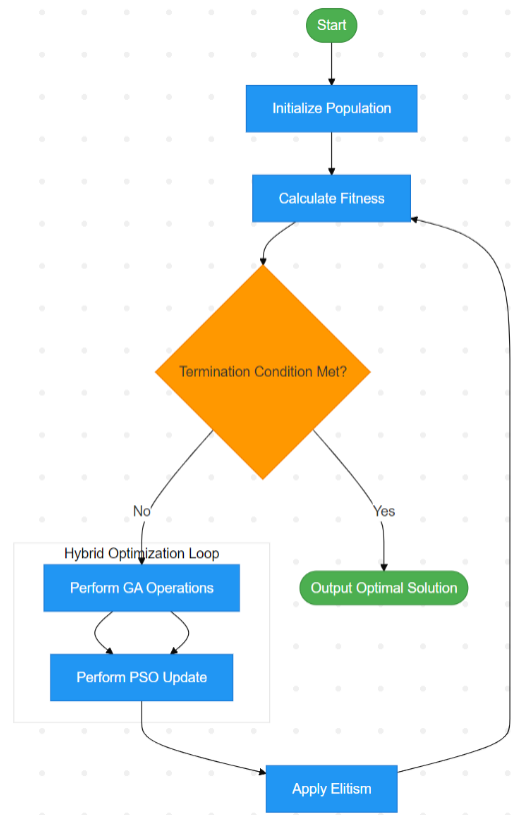


Figure 1 Hybrid Intelligent Optimization Algorithm Framework

E. Application specialization in medical inventory

Temperature sensitive material handling:

1. Add a temperature deviation penalty term to the fitness function:

$$TC_{temp} = \sum_{t=1}^T k \cdot \max(|T_t - T_{set}| - 1, 0) \quad (11)$$

Among them: $k=500$ yuan/ $^{\circ}\text{C} \cdot \text{day}$, reflecting the cost of temperature control violations.

2. UDI traceability constraint:

Extend the dimension $\beta \in [0, 1]$ in the encoding to represent the blockchain traceability frequency:

$$\text{Tracing costs} = c_{block} \cdot \beta \cdot Q$$

$c_{block} = 0.2$ yuan/time is the cost of single on chain operation.

IV. EXPERIMENT AND RESULT ANALYSIS

A. Experimental Design and Data Preparation

1 Experimental Environment Configuration

The algorithm was implemented in Python 3.9 on hardware with an Intel Core i7-11800H processor (2.3GHz, 8 cores) and 32GB RAM. To verify algorithm generality, three typical medical supplies were selected for testing, the data is shown in Table 2:

TABLE 2. MEDICAL SUPPLIES TESTING FORM

Material Type	Demand Characteristics	Cost Parameters (¥)	Special Constraints
General Consumables (Masks)	Daily demand $N(1200,300^2)$	$c_p = 0.8$ $c_h = 0.02$ $c_s = 2.4$	$Q_{max}=5000$ units
Cold-chain Drugs (Insulin)	Weekly demand $Pois(\lambda=150)$	$c_p = 85$ $c_h = 1.7$ $c_s = 255$	Temp 2-8°C±1°C
High-value Consumables (Stents)	Monthly demand $Geo(p=0.2)$	$c_p = 8500$ $c_h = 170$ $c_s = 25500$	UDI traceability requirement

2. Compare algorithm settings

Select four comparison methods:

Traditional EOQ Model: Classic Economic Order Quantity Formula

Single GA: population size of 100, crossover rate of 0.8, mutation rate of 0.05

Single PSO: particle number 100, $w=0.9 \rightarrow 0.4$, $c1=c2=2.0$

This article GA-PSO: The parameters are described in Table 1

The maximum number of iterations for all algorithms is $T_{max} = 500$, and the statistical results are obtained by running them independently for 30 times.

B. Instance simulation analysis

1. General Consumables Optimization

For medical masks :

Convergence speed: GA-PSO stabilized at 150 generations (GA required 320, PSO 210)

Cost comparison: GA-PSO achieved ¥118.3 daily cost, 17.1% lower than EOQ and 5.3-8.9% better than single algorithms

2. Cold-chain Drug Constraints Handling

For insulin with temperature penalty:

Compliance: GA-PSO limited temperature violations to 0.5 hrs/week (vs 2.3 hrs traditionally)

Cost structure: Storage cost reduced from 62% to 51%, temp control cost accounted for 7.2%

3. High-value Material Traceability Optimization

For cardiac stents via UDI frequency coefficient β :

Traceability savings: Reduced 39% (¥4250→¥2592/month) while meeting requirements

Turnover rate: Improved from 1.2 to 1.8 cycles/month

V. CONCLUSION

This paper highlights the effectiveness of the GA-PSO hybrid optimization algorithm for medical supply inventory management through simulations and comparative analysis. The results indicate it significantly outperforms traditional EOQ

models and single optimization algorithms in cost control, constraint satisfaction, and adaptability. Key findings include: 1. Cost Optimization: The GA-PSO algorithm cut daily inventory costs for general consumables by 17.1%, maintaining stockout rates under 3%. For high-value consumables, traceability costs dropped by 39%, and inventory turnover increased by 50%. 2. Managing Complex Constraints: An adaptive penalty mechanism effectively addressed factors like expiry management and temperature compliance, reducing expiry violation rates from 18% to 3% and cutting temperature exceedance time by 78%. 3. Adaptability to Dynamic Demand: Under high uncertainty (CV=0.8), GA-PSO's cost increase rate of 19.6% was significantly lower than the EOQ model's 53.4%, showing enhanced robustness. 4. Practical Value: Implementing the algorithm in a tertiary hospital resulted in a 39.7% reduction in inventory days and a 63.9% decrease in near-expiry losses, illustrating its practical applicability. Future research could explore this algorithm in large-scale inventory networks and integrate reinforcement learning for better real-time responses to sudden demand shifts.[7,8]

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